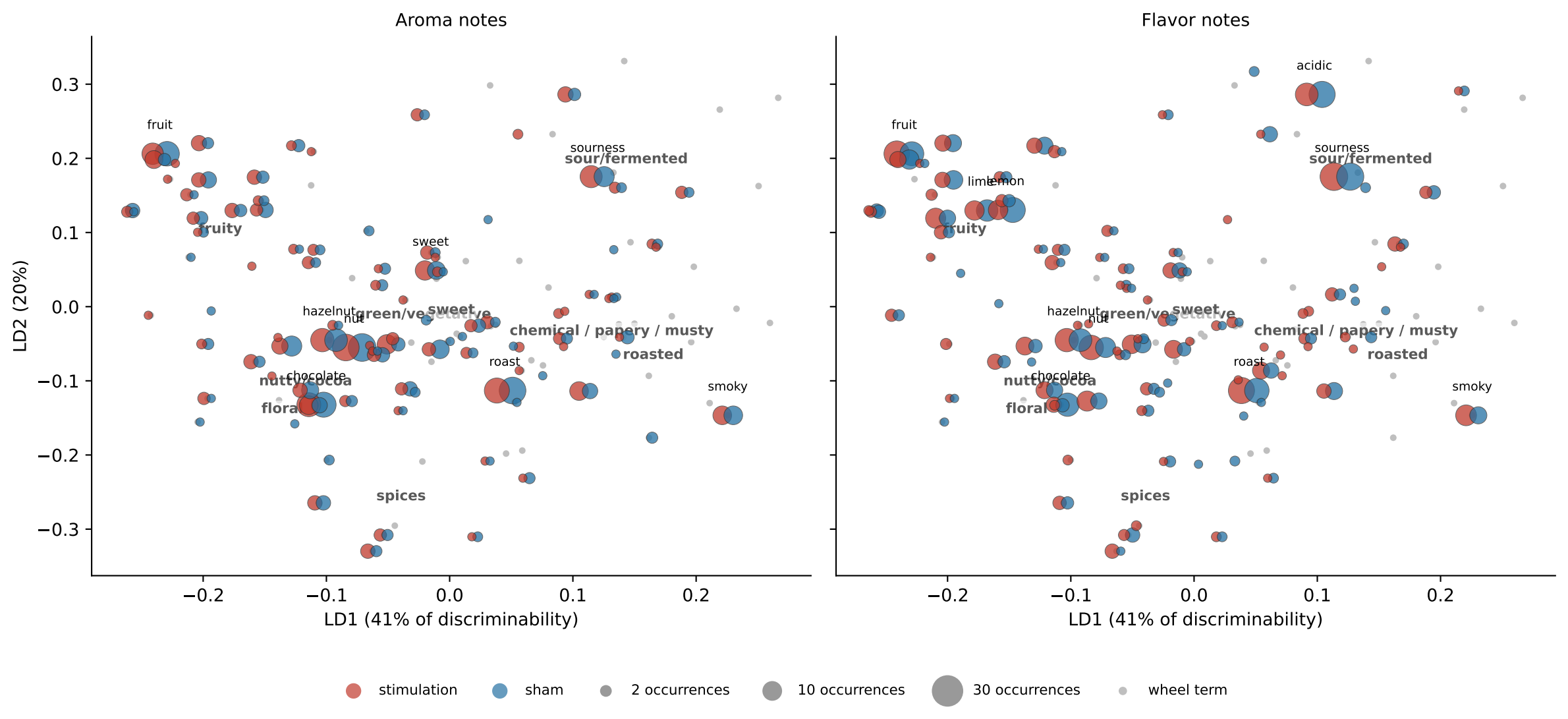


Flavor-wheel space: wheel terms (small grey points, family names in grey) and the words participants used that map to the wheel (coloured points; area = occurrences, red = stimulation, blue = sham, both conditions in the same panel, the two points of a word offset left and right), in the plane (LD1, LD2) that best separates the wheel families (discriminant analysis fitted on the wheel vocabulary only; Qwen3-Embedding-0.6B); green/vegetative, sweet and nutty/cocoa overlap in this plane. Descriptive map; the 2D projection distorts distances.
Tests (family endorsement, paired sign-flip permutation over 9 wheel families; exploratory): aroma omnibus p = 0.018, q = 0.035 (cosine-mapping sensitivity q = 0.057); flavor omnibus p = 0.54, q = 0.54; no single family reaches q < 0.25 (BH over 18 family tests). Differences in the map are not tests of individual words.